

CBSE Class 10 Mathematics - Answer Key

1. 1. The discriminant of the quadratic equation $3x^2 - 4x + 1 = 0$ is

Answer: (C)

Solution:

1. Identify the coefficients a , b , and c from the standard form $ax^2 + bx + c = 0$.
2. The coefficients are $a = 3$, $b = -4$, and $c = 1$.
3. Use the formula for the discriminant $D = b^2 - 4ac$.
4. Substitute the values into the formula: $D = (-4)^2 - 4(3)(1)$.
5. Calculate the result: $16 - 12 = 4$.

Derivation:

$$a = 3, b = -4, c = 1$$

$$D = b^2 - 4ac$$

$$D = (-4)^2 - 4(3)(1)$$

$$D = 16 - 12$$

$$D = 4$$

Quadratic Equations — Nature of Roots